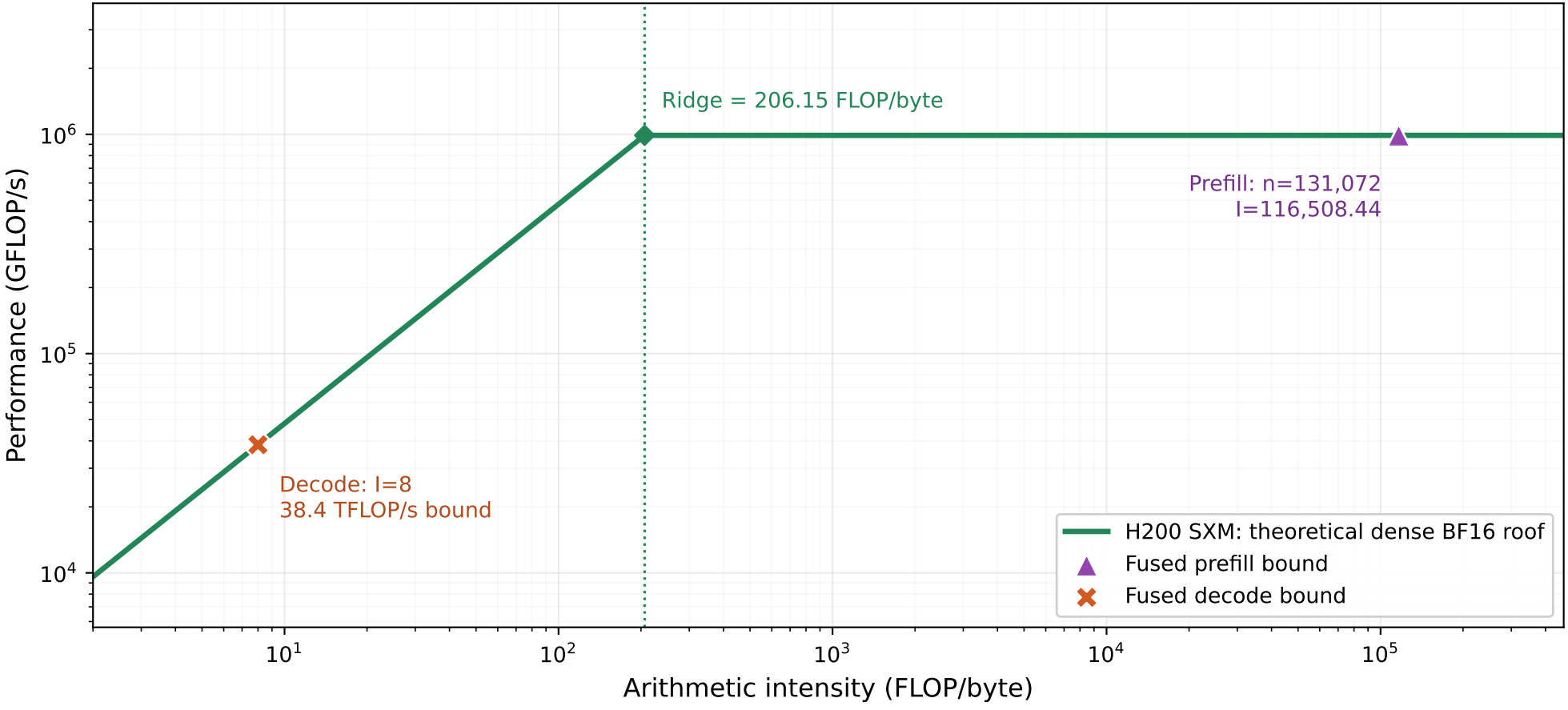
H200 SXM: fused attention roofline

BF16 | query heads = 64 | KV heads = 8 | head dimension = 128



Decode intensity is 25.77 times below the ridge. Attention points are analytical bounds, not measurements.

Default architecture: Llama 3.1 70B at 131,072 tokens. Fused-byte model from assignment section 2.4.

The attention-core model excludes projection/MLP costs and whole-model memory capacity.

H200 specifications: https://www.nvidia.com/en-us/data-center/h200/

Dense peak = 1,979 / 2 = 989.5 TFLOP/s; bandwidth = 4.8 TB/s.